

Observation-Constrained Causal Order and Conformal Cone Geometry

A Four-Gate Closure with Scale Integrability and Cross-Stratum Identifiability

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Abstract

We formulate an observation cone without deriving it from astronomical data or postulating Einstein dynamics. Begin with the universal observational quotient Q_{obs} and equip it with a fibre-compatible observational causal preorder. Its future sets $J_{\text{obs}}^+(x)$ and their boundaries define the observable causal cone. When this ordered set admits a time-oriented Lorentzian manifold realization of dimension greater than two satisfying future and past distinguishability, the Hawking–King–McCarthy–Malament reconstruction results imply that the causal order determines the topology, differentiable structure, and Lorentzian metric up to positive conformal rescaling. Thus the mathematically justified bridge is conditional:

$$(Q_{\text{obs}}, \preceq_{\text{obs}}) + \text{Lorentzian realizability and causality} \implies [g_{\text{obs}}],$$

not an unconditional derivation of spacetime from an arbitrary preorder.

The remaining conformal factor requires scale data. On a multiplicative, zero-memory coherence stratum, $\mathcal{T} = -\log C$ is additive and can serve as an operational duration. We prove, however, that one calibrated worldline does not determine a spacetime-wide conformal factor. Metric selection requires a sufficiently local clock field, an equivalent volume element, or other scale calibration. Astronomy is then assigned its correct role: a remote cross-stratum test of the calibrated null cone. A synthetic redshift–distance law identifies only a ratio such as v_{∂}/H_0 until independent calibration is supplied. The final hostile deterministic attack, seed 20260820, used 180000 samples and passed all 18 registered checks, including conformal invariance, quotient descent, memory violation, scale underdetermination, astronomical degeneracy, and redshift-dependent falsification. The result characterizes causal/conformal observation geometry without claiming the Einstein equations or identifying the cone scale by mathematics alone.

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1 Claim boundary

The objective is to isolate what follows from observation, what follows only after a Lorentzian realization hypothesis, and what remains empirical physics. The paper establishes the ladder

$$\begin{aligned} \text{observable quotient} &\rightarrow \text{observable causal order} \rightarrow \text{conditional conformal realization} \rightarrow \\ &\rightarrow \text{scale calibration} \rightarrow \text{cross-stratum test}. \end{aligned}$$

It does not derive causal order from finite bandwidth, does not derive a smooth manifold from an arbitrary set, and does not derive gravitational dynamics.

2 Observation quotient and causal descent

Let X_{lat} be a class of latent realizations and $\mathfrak{F} : X_{\text{lat}} \rightarrow X_{\text{obs}}$ the declared observation law. Define $x \sim_{\text{obs}} x'$ iff $\mathfrak{F}(x) = \mathfrak{F}(x')$, and let $\pi : X_{\text{lat}} \rightarrow Q_{\text{obs}} = X_{\text{lat}} / \sim_{\text{obs}}$ be the quotient map.

Definition 2.1 (Latent observational reachability). A relation $\preceq_L$ on X_{lat} is an observational reachability preorder when it is reflexive and transitive and when $x \preceq_L y$ means that, under the declared admissible intervention family, an influence initiated at x is observationally distinguishable as capable of reaching y .

Definition 2.2 (Fibre compatibility). The relation $\preceq_L$ is fibre compatible when

$$x \sim_{\text{obs}} x', \quad y \sim_{\text{obs}} y' \implies (x \preceq_L y \iff x' \preceq_L y').$$

Theorem 2.3 (Observational causal quotient). *A latent reachability relation descends by*

$$[x] \preceq_{\text{obs}} [y] \iff x \preceq_L y$$

to a well-defined preorder on Q_{obs} if it is fibre compatible. Conversely, well-defined descent implies fibre compatibility.

Proof. Fibre compatibility makes the definition independent of representatives; reflexivity and transitivity descend immediately. Conversely, if the quotient relation is well defined, replacing either representative cannot alter its truth value, which is fibre compatibility. $\square$

Definition 2.4 (Causal antisymmetrization). For a descended preorder, define

$$q \approx_c q' \iff q \preceq_{\text{obs}} q' \text{ and } q' \preceq_{\text{obs}} q.$$

The causal reduction is

$$Q_c = Q_{\text{obs}} / \approx_c.$$

Proposition 2.5 (Preorder-to-order gate). *The relation induced by $\preceq_{\text{obs}}$ on Q_c is a partial order. If the observational preorder is already antisymmetric, then $Q_c \cong Q_{\text{obs}}$. Thus cycles must either be identified by causal reduction or excluded by a causality hypothesis before Lorentzian reconstruction is invoked.*

Proof. Mutual reachability is an equivalence relation for every preorder. The induced relation on its classes is reflexive and transitive. If two reduced classes are mutually related, their representatives are mutually reachable and hence belong to one $\approx_c$ -class, proving antisymmetry. $\square$

Remark 2.6. The quotient theorem alone does not manufacture causal order. Reachability is additional observational structure that must be specified or recovered from an admissible intervention family. Theorem 2.3 states the exact gate required for that structure to live on the observational quotient.

Definition 2.7 (Observation cone). For $q \in Q_{\text{obs}}$, define

$$J_{\text{obs}}^+(q) = \{q' \in Q_{\text{obs}} : q \preceq_{\text{obs}} q'\}.$$

The notation $\partial J_{\text{obs}}^+(q)$ is introduced only after a topology τ has been supplied. Relative to τ , define

$$\mathcal{C}_{\text{obs}}^+(q) = \partial_{\tau} J_{\text{obs}}^+(q).$$

Proposition 2.8 (Bare order does not yet define a cone boundary). *A preorder by itself does not determine the set-theoretic boundary of its future sets. The observation cone therefore requires either an independently justified compatible topology or the topology recovered under the Lorentzian realization gate below.*

Proof. Boundary is a topological operation. The same ordered set equipped with different topologies can give different closures, interiors, and boundaries of the same future set. In particular, the discrete topology makes every set open and closed and hence gives empty boundary. $\square$

3 From causal order to conformal geometry

An arbitrary preorder need not be a Lorentzian spacetime. The established reconstruction theorem applies only after explicit realizability, dimensional, and causality assumptions.

Assumption 3.1 (Lorentzian realization gate). The ordered observational quotient $(Q_{\text{obs}}, \preceq_{\text{obs}})$ is isomorphic to the causal order of a connected, time-oriented Lorentzian manifold (M, g) of dimension $n > 2$. The realization is future and past distinguishing. The order is understood together with the time orientation required to distinguish future from past.

Theorem 3.2 (Conditional causal reconstruction). *Under Assumption 3.1, the observational causal order determines the Lorentzian conformal geometry: any order isomorphism to another distinguishing Lorentzian realization is a conformal diffeomorphism. Hence the observation structure determines a conformal class*

$$[g_{\text{obs}}] = \{\Omega^2 g : \Omega > 0\}.$$

Proof. This is the Hawking–King–McCarthy–Malament reconstruction result specialized to the observational realization: in the relevant distinguishing causality class, causal structure determines the manifold/conformal structure up to conformal isometry [1, 2, 3]. $\square$

Corollary 3.3 (Conformal invariance of the observation cone). *Positive rescaling $g \mapsto \Omega^2 g$ preserves causal vectors, causal order, and null boundaries. Therefore $\mathcal{C}_{\text{obs}}^+$ is conformal data and does not determine absolute metric scale.*

Proof. For every tangent vector v , $(\Omega^2 g)(v, v) = \Omega^2 g(v, v)$ has the same sign because $\Omega > 0$. Time orientation fixed, causal futures and their null boundaries are unchanged. $\square$

Non-claim 3.4. Theorem 3.2 is an established Lorentzian reconstruction theorem, not a new theorem of Resolution Geometry. The contribution here is the typed assembly: observational quotient, fibre-compatible reachability, explicit realization gate, and separation of conformal reconstruction from scale and dynamics.

4 Operational duration and the scale lock

Let $C_{ij} \in (0, 1]$ be a measured coherence assigned to a composable segment. Define

$$\mathcal{T}_{ij} = -\log C_{ij}.$$

Proposition 4.1 (Additive zero-memory clock). *If coherence composition is multiplicative, $C_{02} = C_{01}C_{12}$, then*

$$\mathcal{T}_{02} = \mathcal{T}_{01} + \mathcal{T}_{12}.$$

If instead

$$\mathcal{T}_{02} = \mathcal{T}_{01} + \mathcal{T}_{12} + \mathcal{M}_{012},$$

strict additivity holds exactly on the stratum $\mathcal{M}_{012} = 0$.

Proof. Apply $-\log$ to multiplication. The second statement follows by subtracting the two additive terms. $\square$

Additivity creates an operational duration, but this alone does not globally select a representative of $[g]$.

Theorem 4.2 (Single-worldline scale underdetermination). *Calibration of proper duration along one timelike curve does not, in general, determine the conformal factor on the surrounding spacetime.*

Proof. In coordinates (t, x) near the curve $x = 0$, take $g_1 = g$ and $g_2 = \Omega^2 g$ with

$$\Omega(t, x) = 1 + \frac{x^2}{4}.$$

The two metrics agree along $x = 0$ and therefore generate the same calibrated clock there, but differ for $x \neq 0$. Both belong to the same conformal class. $\square$

Corollary 4.3 (Sufficient scale data). *A representative metric can be selected when the operational calibration supplies the conformal factor locally throughout the region of interest, or when equivalent local scale data such as a compatible volume element are given. The selected metric is unique within the conformal class wherever that local scale datum is fixed.*

Proof. If $g' = \Omega^2 g$, pointwise scale data determine the positive scalar Ω . A volume form in dimension n transforms as $d\mu_{g'} = \Omega^n d\mu_g$ and therefore also fixes Ω pointwise. $\square$

Remark 4.4 (Precise role of the coherence clock). The clock is a candidate scale datum only after calibration and locality are established. Proposition 4.1 proves composition on the zero-memory stratum; it does not prove that $\mathcal{T}$ equals GR proper time or that one clock fixes an entire metric field.

5 Scale integrability and atlas consistency

Local duration calibration is not sufficient unless independently calibrated patches glue to one positive conformal factor. Let $\omega_\alpha = \log \Omega_\alpha$ be local log-scale representatives on patches U_α .

Definition 5.1 (Scale-integrability gate). A local clock atlas is scale integrable when the differences $\omega_\beta - \omega_\alpha$ agree on overlaps and have zero accumulated increment around every closed overlap chain. Equivalently, the induced log-scale cocycle is exact.

Theorem 5.2 (Metric representative from an integrable local scale field). *Let $[g]$ be fixed on a connected region and suppose calibrated local scale data form an exact positive conformal cocycle. Then they define a global positive function Ω and hence a representative $g_{\text{cal}} = \Omega^2 g$. The representative is unique after fixing one overall calibration convention. If the overlap cocycle has nonzero holonomy, no single global representative realizes all local clock calibrations.*

Proof. Exactness gives local potentials that agree after the declared transition adjustments, so they glue to a global ω . Set $\Omega = e^\omega > 0$. Two glued solutions have locally constant log-ratio; connectedness and one calibration convention remove that constant. Nonzero loop sum contradicts the existence of a single-valued potential. $\square$

Remark 5.3. This is the additional closure forced by attack: additivity along composable segments and pointwise calibration are necessary but not enough. Global metric selection also requires overlap and path consistency.

6 Astronomy as cross-stratum validation

Astronomy is not used to generate the cone. After local reconstruction and scale calibration, remote null propagation tests whether the same cone structure is realized across observational strata.

Let a simplified redshift–distance family have the form

$$d(z) = \frac{v_\partial}{H_0} f(z; \vartheta),$$

where ϑ denotes other declared dimensionless parameters.

Proposition 6.1 (Astronomical scale degeneracy). *With f and ϑ fixed, the redshift–distance law identifies the ratio v_∂/H_0 , not v_∂ and H_0 separately. Independent calibration of either factor is required to identify the other.*

Proof. For every $a > 0$, replacing (v_∂, H_0) by (av_∂, aH_0) leaves $d(z)$ unchanged. $\square$

Definition 6.2 (Cross-stratum cone invariance test). For admissible strata α , compare their reconstructed conformal structures and, after independent scale calibration, their boundary speeds:

$$[g_\alpha] \stackrel{?}{=} [g_{\text{obs}}], \quad v_\partial^{(\alpha)} \stackrel{?}{=} v_\partial.$$

A redshift-dependent alternative $v_\partial(z)$ is tested against the constant boundary model by its structured residual across z . The comparison is defined only after specifying a correspondence map between local and remote observable structures.

Proposition 6.3 (Correspondence and nuisance gates). *Cross-stratum equality is meaningful only relative to a declared identification of observables, events, or invariant summaries between strata. Moreover, a redshift-dependent cone violation is identifiable only against a restricted and reported nuisance family: an unrestricted nuisance term with the same functional shape can absorb the violation exactly.*

Proof. Without a correspondence, equality of labelled structures is undefined; equal marginal distributions need not give eventwise or structural identity. For the second claim, if a nuisance basis contains the candidate violation function, its coefficient can reproduce that function and leave zero residual. $\square$

Remark 6.4. The identification $v_{\partial} = c$ is empirical physics: calibrated optical and astronomical null propagation identify the reconstructed boundary with the experimentally established light cone. It is not a consequence of quotient universality or causal reconstruction alone.

7 Four-lock closure theorem

Theorem 7.1 (Observation-cone architecture). *Subject to the hypotheses stated in the preceding sections, the observation-cone construction separates into four locks:*

1. Order lock: *observational equivalence produces Q_{obs} ; fibre-compatible reachability descends; mutual-reachability cycles are reduced or excluded by causality.*
2. Conformal lock: *compatible topology, Lorentzian realizability, time orientation, dimension $n > 2$, and distinguishing causality permit causal-order reconstruction of $[g_{\text{obs}}]$.*
3. Scale lock: *$-\log C$ is additive on the zero-memory stratum, but metric selection requires local calibration plus exact overlap/path consistency, or equivalent local volume data.*
4. Cross-stratum lock: *astronomy tests remote null consistency only after a correspondence map, independent scale calibration, and a declared nuisance model are fixed.*

None of these locks entails Einstein dynamics.

Proof. Item 1 is Theorem 2.3; item 2 is Theorem 3.2; item 3 follows from Proposition 4.1 and Theorem 4.2; item 4 is the operational test following Proposition 6.1. The Einstein equations require additional dynamical fields and laws absent from these premises. $\square$

8 Final hostile structure attack

The analytic results do not depend on simulation. The final battery attacks the gates and typing assumptions. It used seed 20260820, 180000 randomized samples, and passed all 18 registered checks.

Attack	Result	Closure forced by attack
Fibre-compatible descent	pass	Relation is constant on both fibres.
Hostile representative change	pass	Non-descending relation rejected.
Preorder cycle	pass	Mutual reachability detected and reduced.
Conformal causal type/order	pass	No mismatches under positive rescaling.
Signed scale	pass	Nonpositive factor rejected.
Bare-order boundary	pass	Topology gate is necessary.
Zero-memory clock	pass	Additivity residual 4.44×10^{-16} .
Memory attack	pass	Mean defect 0.19919.
Exact scale atlas	pass	Loop residual 2.22×10^{-16} .
Inconsistent scale atlas	pass	Holonomy RMS 0.02006.
Single-worldline scale	pass	Transverse gap 0.43826.
Astronomical ratio	pass	Exact v_∂/H_0 degeneracy.
Speed-trend alternative	pass	Restricted-model residual 0.00598.
Nuisance absorption	pass	Matching nuisance residual 6.05×10^{-16} .
Cross-stratum scrambling	pass	Correspondence-sensitive RMS 28.1594.
Independent calibration	pass	Relative bias 2.83×10^{-5} .

The successful outcome does not mean that the strongest informal ladder survived unchanged. The attack inserted four mandatory gates: causal antisymmetrization, compatible topology, scale-cocycle exactness, and cross-stratum correspondence with nuisance control. With those gates explicit, no tested internal contradiction remains.

9 Reproducibility record

The final artifacts are

- `observation_cone_structure_attack.py`;
- `observation_cone_structure_attack.json`.

The JSON records the seed, sample count, thresholds, all check outcomes, and the scope verdict. The astronomy component is synthetic and tests identifiability; it is not a fit to observational catalogues. All numerical checks are metadata and are not premises of Theorem 7.1.

10 Non-claims and conclusion

Non-claim 10.1. This paper does not prove that every observational preorder has a Lorentzian realization, does not derive smooth manifold structure from arbitrary records, does not prove that coherence duration equals proper time, does not derive the Einstein equations or a stress tensor, and does not estimate c or H_0 from real astronomical data.

The defensible result is both strong and bounded. Observation supplies a universal quotient. An additional fibre-compatible reachability structure can supply observational causal order. Under explicit Lorentzian realizability and causality gates, established reconstruction mathematics turns that order into a conformal cone geometry. Operational duration can select scale only when it is additive, sufficiently local, and globally integrable across overlaps. Astronomy then tests whether the same calibrated null structure survives remotely, relative to

a declared cross-stratum correspondence and nuisance model; it does not create the cone or independently determine absolute scale.

Observable distinguishability plus admissible reachability yields an observational causal order.

A distinguishing Lorentzian realization turns that order into conformal cone geometry.

Local operational calibration selects metric scale.

Astronomy tests cross-stratum invariance.

Gravitational dynamics remains additional physics.

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